



# Proton pump inhibitors associated with an increased risk of mortality in elderly: a systematic review and meta-analysis

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## Abstract

**Purpose** The increased use of proton pump inhibitors (PPIs) in the elderly has raised concerns about potential severe adverse effects. Our systematic review investigated the mortality associated with PPI use in elderly populations.

**Methods** We searched MEDLINE, EMBASE, and the Cochrane Library for relevant publications until August 2022. We included randomized controlled trials (RCTs), quasi-RCTs, and observational studies on the association between proton pump inhibitors and mortality in the elderly. To estimate the pooled relative risk (RR) and 95% confidence interval (CI), the inverse-variance random effect model was used. Heterogeneity was assessed using the  $I^2$  test. Subgroup analyses were performed by follow-up period, population, and study design.

**Results** A total of 4 RCTs and 36 cohort studies were included in the meta-analysis. Four RCTs showed that there was no significant association between PPIs and the risk of death. From 23 observational studies (26 cohorts), the use of proton pump inhibitors was not significantly associated with increased mortality in the elderly (RR 1.14; 95% CI, 0.90–1.45). However, when controlling for covariates from 33 observational studies (41 cohorts), proton pump inhibitors in older adults aged 50 years or more were significantly associated with a 15% higher risk of mortality compared to nonusers (RR 1.15; 95% CI, 1.10–1.20).

**Conclusions** Our meta-analysis of RCTs found that PPIs did not show a significant association with increased mortality risk in older adults. However, the meta-analysis of cohort studies and long-term follow-up studies showed a higher increased risk of death with PPI use in older adults. The prescription of PPIs in patients aged 50 years or older should be carefully considered.

**Keywords** Proton pump inhibitors · Mortality · Elderly · Systematic review · Meta-analysis

## Introduction

In recent decades, gastrointestinal diseases have been on the rise in older populations [1]. In this demographic, proton pump inhibitors (PPIs) are commonly used to treat heartburn, gastroesophageal reflux disease, peptic ulcer disease, and *Helicobacter pylori* infections [2, 3]. In an Irish study among adults aged 40 years or older with intellectual disabilities, the largest population of PPI users was aged 50–64 years (53.4%), and 30.2% were aged 65 years or older [4].

Concerns of potentially severe adverse effects of PPIs have increased, particularly in older adults. A recent review demonstrated that the use of PPI in older adults was associated with an increased risk of osteoporotic-related fractures, *Clostridium difficile* infection, community-acquired pneumonia, vitamin B12 deficiency, kidney disease, and dementia [5]. These potential side effects are of particular concern to the elderly because this vulnerable population is already

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## Key points

- Little is known about the excess risk of death associated with proton pump inhibitor use in the elderly.
- We conducted a systematic review and meta-analysis of randomized controlled trials and observational studies regarding the association between proton pump inhibitors and the risk of death in adults aged 50 years or older.
- Our meta-analysis found a 15% increased risk of death in elderly people with PPI use compared to nonusers. Longer follow-up and cohort studies revealed an increased risk of death among older patients with cancers, cardiovascular disease, and kidney disease.
- Awareness of the increased mortality with PPI use should be raised, and the need to limit PPI prescriptions to the elderly where the benefits outweigh the potential risks should be emphasized.

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more likely to suffer from an increased risk of these conditions and consequently, severe morbidity. As a result, the guidelines recommend avoiding the use of PPIs for longer than 8 weeks in the elderly except in high-risk patients and discontinuing or reducing PPIs in older adults with more than 8-week usage of PPIs for uncomplicated peptic ulcer disease or erosive peptic esophagitis [5–7].

Additionally, recent studies have reported that PPIs were associated with increased mortality in the general population and/or patients with cancer [8–11]. The association between PPIs and mortality in the elderly population is undefined, but it has been shown that they are potentially affected [10, 11]. Since previous studies have yielded inconsistent results for mortality in older adults who take PPIs and the excess risk of death associated with PPI use has not been systematically investigated, pooled estimates that combined relative risks for mortality from each study are needed. Thus, we performed a systematic review and meta-analysis of the association between PPIs and mortality in older adults.

## Methods

### Literature search and study selection

We conducted searches on MEDLINE, EMBASE, and the Cochrane Library for articles published up to August 24, 2022. To search for relevant articles, we used MeSH terms and text words related to outcome, such as “mortality,” “death,” and “fatality,” and intervention-related search terms, including the drug, brand, and chemical names of “proton pump inhibitor” (i.e., benatoprazole, dextansoprazole, esomeprazole, lansoprazole, omeprazole, pantoprazole, rabeprazole, and tenatoprazole). The search strategy is presented in Supplement Table 1.

All studies that met the following criteria were included: (1) investigated the association between PPIs and the risk of mortality in adults aged 50 years or older; (2) compared PPIs and control such as placebo or active comparator; (3) reported the quantified relative risk of death; (4) conducted randomized controlled trials (RCTs), non-randomized controlled studies, and observational studies; and (5) written in English. If the participants’ age was not clearly mentioned in the inclusion criteria for the study, we included studies with participants of both a mean or median  $\geq 66$  years of age and an interquartile range of  $\geq 56$  years. Two independent reviewers examined the study selection, and the third reviewer was consulted when there was a disagreement. We registered the study protocol in PROSPERO (CRD42020179631) and followed the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines [12].

### Quality assessment

Two independent researchers conducted a quality assessment of the included studies. For RCTs, we used version 2 of the Cochrane Risk of Bias (ROB 2) tool, which is composed of five items (bias arising from the randomization process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result), to assess the quality of studies with three levels (low, some concerns, and high risk of bias) [13]. The Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) was used to evaluate the quality of non-RCTs. ROBINS-I consists of five items from three domains: pre-intervention, at-intervention, and post-intervention domains [14]. Pre-intervention domain includes bias due to confounding and bias in selection of participants into the study. At-intervention domain includes biases in classification of intervention, and post-intervention domain includes bias due to deviations from intended interventions, bias due to missing data, bias in measurement of the outcome, and bias in selection of the reported result. Four levels were used to assess the risk of bias in included studies: low, moderate, serious, and critical.

### Data extraction

We collected information on the study and demographic characteristics (country, study design, data source, study period, mean age, percentage of male participants), exposure (definition and number of PPI users and controls, follow-up period), and outcomes (number of death of PPIs and comparators, relative risk [RR] of mortality) from the included studies. The confounding variables were extracted in regression analysis when applicable.

The primary outcomes were the unadjusted and adjusted estimates of the risk of mortality associated with PPI use. We used the best-adjusted relative risks with a 95% confidence interval (CI) after controlling the confounding variables from each included study for the meta-analysis.

### Statistical analysis

To estimate the pooled RRs with CIs, the inverse-variance random effect model was used. We calculated log RR and standard error (SE) using the 95% CIs or *P*-values reported in the studies. Each study reported a different type of relative risk, such as RR, hazard ratio (HR), or odds ratio (OR). In our meta-analysis, HRs were considered RRs [15, 16], and ORs were converted to RRs using the method described by Zhang and Yu, which uses OR and the incidence of mortality in the control group [17]. The studies that reported OR

**Table 1** Study characteristics of included studies

| Author, year                        | Country                                                | Study design | Data source/study period                                                                      | Definition and number of participants (PPIs/nonusers)                                                                                                                                                                                                         | Mean age (SD)/male (%)                                                                    | Exposure/co-medication            | Follow-up | Numbers of death                                        | Adjustments (matching variables in case-control studies)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Mortality (95% CI)                                                                                       |
|-------------------------------------|--------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------|-----------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Randomized controlled trials</b> |                                                        |              |                                                                                               |                                                                                                                                                                                                                                                               |                                                                                           |                                   |           |                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                          |
| Hasselgren et al., 1997 [31]        | Sweden, Norway                                         | RCT          | 29 centers/NR                                                                                 | Patients aged $\geq 60$ years with peptic ulcer bleeding 322 (159/163)                                                                                                                                                                                        | Omeprazole 74.5 (8.2), placebo 74.3 (7.4)/58.1%                                           | Omeprazole                        | 21 days   | Omeprazole 11/159 (6.9%), placebo 1/163 (0.6%)          | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Omeprazole vs placebo, $P=0.012$                                                                         |
| Marker et al., 2019 [39]            | Denmark, Finland, Norway, Switzerland, Netherlands, UK | RCT          | International Stress Ulcer Prophylaxis in the Intensive Care Unit (ISP-ICU) trial 2016–2017   | Patients at risk for gastrointestinal bleeding 3261 (1635/1626)                                                                                                                                                                                               | Median 67 (IQR 56–75)/64%                                                                 | Pantoprazole                      | 1 year    | Pantoprazole 610/1635 (37.3%), placebo 601/1626 (37.0%) | Age, type of admission, sepsis-related organ failure assessment score, hematological malignancy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | aRR 0.99 (0.90–1.09)                                                                                     |
| Moayyedi et al., 2019 [40]          | 580 centers in 33 countries                            | RCT          | Cardiovascular Outcomes for People Using Anticoagulation Strategies (COMPASS) trial/2013–2016 | Patients aged $\geq 65$ years with stable coronary or peripheral arterial disease 17,598 (879/8807)                                                                                                                                                           | Pantoprazole 67.6 (8.1), placebo 67.7 (8.1)/78.4%                                         | Pantoprazole                      | 3 years   | PPI 630/8791 (7.2%), placebo 614/8807 (7.0%)            | NR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | HR 1.03 (0.92–1.15)                                                                                      |
| Wu et al., 2011 [56]                | China                                                  | RCT          | Three cardiology centers/2008–2010                                                            | Patients with acute coronary syndromes and high risk for GI bleeding 665 (333/332)                                                                                                                                                                            | NR/73.7%                                                                                  | Pantoprazole                      | 30 days   | Pantoprazole 35/333 (10.5%), placebo 34/332 (10.2%)     | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Pantoprazole vs placebo, $P=1.00$                                                                        |
| <b>Observational studies</b>        |                                                        |              |                                                                                               |                                                                                                                                                                                                                                                               |                                                                                           |                                   |           |                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                          |
| Adelborg et al., 2018 [18]          | Denmark                                                | Cohort       | Danish medical registries/1995–2014                                                           | Patients with heart failure 46,198 (42,902/3296)                                                                                                                                                                                                              | PPI median 78 (IQR 69–85)/54%<br>H2RA median 76 (IQR 68–83)/52%                           | PPIs<br>H2RA                      | 5 years   | PPI 25,835/42,902 (76.8%), H2RA 1964/3296 (60.7%)       | Age, index year, sex, time from heart failure hospital admission date until first prescription of a H2RA or PPI, CAD, valvular heart disease, hypertension, atrial fibrillation, atrial flutter, VTE, stroke, intermittent claudication, diabetes, obesity, cancer, chronic pulmonary disease, CKD, dementia, depression, illicit drug abuse, alcohol abuse, smoking, anemia, PUD, GERD, liver disease, alcohol disorders, musculoskeletal disorders, inflammatory bowel disease, beta blockers, ACEI/ARB, diuretics, statins, NSAID, antithrombotics, benzodiazepines, opioids, gross income, employment | NR                                                                                                       |
| Ayyagari et al., 2020               | USA                                                    | Cohort       | VA Central Cancer Registry/2002–2016                                                          | 1. Patients with esophageal adenocarcinoma 6530 (3865/2665)<br>2. Patients with esophageal squamous cell carcinoma 3227 (1466/1761)<br>3. Patients with cardia gastric cancer 2361 (1361/1000)<br>4. Patients with non-cardia gastric cancer 3142 (1818/1324) | 1. 67.1 (9.2)/99.5%<br>2. 67.0 (9.6)/98.7%<br>3. 68.1 (9.8)/98.6%<br>4. 69.6 (11.0)/97.5% | PPIs<br>H2RA                      | NR        | NR                                                      | Age, sex, race, BMI, alcohol use, smoking status, stage, grade, chemotherapy, radiation, surgery, pre-diagnosis PPI use, post-diagnosis aspirin/NSAID use, post-diagnosis statin use<br>GNCC only: H. pylori infection                                                                                                                                                                                                                                                                                                                                                                                    | 1. aHR 0.97 (0.91–1.03)<br>2. aHR 1.05 (0.98–1.14)<br>3. aHR 1.13 (1.03–1.24)<br>4. aHR 1.04 (0.96–1.13) |
| Baek et al., 2022 [20]              | South Korea                                            | Cohort       | National Health Insurance Service/2017–2018                                                   | Patients treated with advanced NSCLC 955                                                                                                                                                                                                                      | NR                                                                                        | PPIs/immune checkpoint inhibitors | NR        | NR                                                      | Age, sex, respiratory disease, viral hepatitis, antibiotics, corticosteroid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 65–74 years: aHR 1.12 (0.91–1.36)<br>$\geq 75$ years: aHR 1.82 (1.34–2.47)                               |

Table 1 (continued)

| Author, year                | Country      | Study design | Data source/study period                                                                                            | Definition and number of participants (PPIs/nonusers)                                                   | Mean age (SD)/male                                                         | Exposure/co-medication | Follow-up            | Numbers of death                                                                                           | Adjustments (matching variables in case-control studies)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Mortality (95% CI)                                       |
|-----------------------------|--------------|--------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------|----------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Baik et al., 2022 [21]      | USA          | Cohort       | Medicare Prescription Drug (Part D) enrollees/2007–2017                                                             | Elderly Medicare fee-for-service beneficiaries 1,930,728 (336,619/1,522,073)                            | NR/42.4%                                                                   | PPIs<br>H2RA           | Median 3.8 years     | PPI 25,774/<br>336,619 (7.7%),<br>nonuser 45,346/<br>1,462,216 (3.1%)                                      | Age, gender, race, living area, AML, atrial fibrillation, cataract, CKD, COPD, heart failure, diabetes, glaucoma, hip/pelvic fracture, IHD, depression, Alzheimer's disease, dementia, osteoporosis, rheumatoid arthritis/osteoarthritis, stroke/transient ischemic attack, breast cancer, colorectal cancer, prostate cancer, lung cancer, endometrial cancer, anemia, asthma, hyperlipidemia, hypertension, hyperplasia, hypothyroidism, alcohol use disorders, anxiety disorders, bipolar disorder, drug use disorder, personality disorders, schizophrenia, epilepsy, cystic fibrosis, fibromyalgia, viral hepatitis (general), liver disease cirrhosis, leukemias and lymphomas, migraine, mobility impairments, obesity, opioid use disorder, PVD, tobacco use disorders, pressure ulcers, deafness | aHR 1.10<br>(1.08–1.11)                                  |
| Bakhshwin et al., 2022 [22] | Saudi Arabia | Cohort       | King Faisal Medical Complex/2020                                                                                    | COVID-19 hospitalization patients aged ≥ 60 years 145                                                   | 69.2 (8.1)/55.2%                                                           | PPIs                   | NR                   | PPI 12/102<br>(11.8%), nonuser<br>4/43 (9.3%)                                                              | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | NR                                                       |
| Bell et al., 2010 [23]      | Finland      | Cohort       | 1. Long-term care wards<br>2. Acute geriatric wards, nursing homes/1999–2000                                        | Institutionalized older people 1,1004 (231/773)<br>2,425 (91/334)                                       | 1, 81.3 (10.9)/25.5%<br>2, 96.1 (7.0)/18.4%                                | PPIs                   | 12 months            | 1. PPI 77/231<br>(33.3%), nonuser<br>206/773 (26.6%)<br>2. PPI 33/91<br>(36.3%), nonuser<br>73/334 (21.9%) | 1. Age, sex, CCI, malnutrition<br>2. Age, sex, CCI, delirium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1. aHR 1.37<br>(1.05–1.78)<br>2. aHR 1.82<br>(1.20–2.78) |
| Bradley et al., 2019 [24]   | USA          | Cohort       | University of Massachusetts Memorial Health Care System Therapeutic Clinical Surveillance Software System/2012–2014 | Elderly patients with <i>Clostridium difficile</i> infection (> 65 years of age) 874 (320/554)          | Survivors 76.5 (8.1), died 78.4 (8.2)/41.6%                                | PPIs                   | 180 days             | PPI 24/320 (7.5%),<br>nonuser 85/554<br>(15.3%)                                                            | Age, Hispanic, African American, Nursing home, CCI, severe infection, hospital acquired, bacterial other, probiotic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | aHR 0.45<br>(0.28–0.72)                                  |
| Brozek et al., 2017 [25]    | Austria      | Cohort       | 13 Austrian social insurance authorities/2008–2011                                                                  | Hip fracture patients ≥ 50 years without anti-osteoporotic drugs 15,958 (11,390/4568)                   | NR<br>PPIs 32.1%                                                           | PPIs                   | 3 years              | NR                                                                                                         | Age, gender                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | aHR 1.37<br>(1.17–1.60)                                  |
| Cetin et al., 2020 [26]     | Austria      | Cohort       | 13 Austrian statutory sickness funds/2005–2016                                                                      | 1. Patients with dementia 28,428 (12,979/15,449)<br>2. Patients without dementia 56,856 (21,948/34,908) | 1. Median 82.2 (IQR 76.8–86.6)/32.2%<br>2. Median 82.8 (IQR 77.9–86.5)/30% | PPIs                   | ≥ 4 years            | 1. 19,267/28,428<br>(67.8%)<br>2. 20,484/56,856<br>(36.0%)                                                 | Age, sex, NSAID, the number of distinct drug classes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1. aHR 1.03<br>(1.01–1.05)<br>2. aHR 1.15<br>(1.05–1.26) |
| Cholin et al., 2021 [27]    | USA          | Cohort       | Chronic kidney disease registry/2007–2017                                                                           | Patients with chronic kidney disease 25,455 (8646/15,961)                                               | PPI: 72.8 (10.9)/38.4%<br>Nonuser 73.4<br>(11.1)/44.4%                     | PPIs                   | Median<br>4.14 years | PPIs 1845/8646<br>(21.3%), nonuser<br>3521/15,961<br>(22.1%)                                               | Age, race, sex, eGFR, BMI group, hemoglobin, potassium, CO <sub>2</sub> , diabetes, hypertension, CVD, PVD, CAD, CHF, malignancy, ACE/ARB, beta blockers, smoking and insurance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | aHR 0.97<br>(0.91–1.03)                                  |

**Table 1** (continued)

| Author, year                   | Country           | Study design | Data source/study period                                                                                                                                                                          | Definition and number of participants (PPIs/nonusers)                                    | Mean age (SD)/male                                         | Exposure/co-medication                                            | Follow-up            | Numbers of death                                       | Adjustments (matching variables in case-control studies)                                                                                                                                                                                                                                                                                                                                       | Mortality (95% CI)         |
|--------------------------------|-------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------|----------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| De Francisco et al., 2017 [28] | Spain             | Cohort       | Fresenius Medical Care NephroCare dialysis clinics/2014                                                                                                                                           | Hemodialysis patients 2242 (1776/466)                                                    | 68 (56–76)/62.8%                                           | PPIs                                                              | 22.8 months          | PPIs 433/1776 (24.4%), nonuser 82/466 (15.6%)          | Age, sex, dialysis vintage, diabetes mellitus, CCI, hemodialysis clinical parameters, laboratory values, antithrombotic agents                                                                                                                                                                                                                                                                 | aHR 1.37, 95% CI 1.05–1.78 |
| Farhat et al., 2020 [29]       | Canada            | Case-control | Cerner Health Facts database/2000–2016                                                                                                                                                            | Patients with myocardial infarction and started clopidogrel treatment 16,823 (9301/7522) | Case 70.2 (12.1)/56.3%<br>Control 70.4 (11.7)/56.7%        | PPIs                                                              | 12 months            | NR                                                     | Demographics, smoking, comorbidities, use of comedications, prior cardiovascular procedures, and hospital characteristics                                                                                                                                                                                                                                                                      | aOR 1.04 (0.95–1.15)       |
| Hálfáðnason et al., 2020 [30]  | Iceland           | Cohort       | Icelandic Cancer Registry, Icelandic Medicines Registry, Icelandic Population Register, Cause of Death Register, Landspítali/2007–2012                                                            | Patients with prostate cancer 1058 (347/711)                                             | PPI median 69 (IQR 63–76)<br>Nonuser median 69 (IQR 62–75) | PPIs                                                              | Median 4.6 years     | PPI 59/347 (17.0%), nonuser 144/711 (20.3%)            | Age, calendar period, clinical stage, Gleason score, medication-based comorbidity, surgery, endocrine and/or chemotherapy, radiotherapy                                                                                                                                                                                                                                                        | aHR 1.02 (0.73–1.43)       |
| Jun et al., 2021 [32]          | USA, Europe, Asia | Cohort       | 11 tertiary referral centers/2017–2019                                                                                                                                                            | HCC patients treated with immunotherapy 314 (85/229)                                     | 66 (IQR 59–72)/79%                                         | PPIs                                                              | Median 9.2 months    | NR                                                     | -                                                                                                                                                                                                                                                                                                                                                                                              | HR 1.14 (0.84–1.54)        |
| Juurink et al., 2011 [33]      | Canada            | Case-control | Canadian Institute for Health Information Discharge Abstract Database, Ontario Public Drug Program Benefit Program, Ontario Health Insurance Plan database, Registered Persons Database/2002–2008 | Older adults ≥ 66 years treated with clopidogrel after stroke 300 (96/205)               | Median 78 (IQR 73–83)/46.8%                                | PPIs                                                              | Up to 78 months      | PPI 22/78 (28.2%), nonuser 35/205 (17.1%)              | Age, gender, income quintile, CCI, length of stay in the hospital, hospital type, history of diabetes, stroke, coronary revascularization, ACEI, ARB, aspirin, adrenergic receptor antagonists, CCB, statins, lipid-lowering agents, diuretics, warfarin, cytochrome P-450 2C19 inhibitors/inducers, cytochrome P-450 3A4 inhibitors/inducers (age, gender, stroke, transient ischemic attack) | aOR 1.84 (0.88–3.89)       |
| Liabeuf et al., 2021 [34]      | France            | Cohort       | CKD-REIN/2013–2016                                                                                                                                                                                | Patients with stage 3 or 4 chronic kidney diseases 3023 (981/2042)                       | PPI 70 (IQR 64–77), no PPI 68 (IQR 58–76)/65.5%            | PPIs                                                              | PPI median 3.9 years | PPI 49/981 (5.0%), nonuser 167/2042 (8.2%)             | Age, sex, education, diabetes, estimated glomerular filtration rate, BMI, smoking, albuminemia, anemia, systolic blood pressure, history of acute kidney injury, CVD                                                                                                                                                                                                                           | aHR 2.42 (1.73–3.39)       |
| Lo et al., 2022 [35]           | USA               | Cohort       | Nurses' Health Survey, Health Professionals/2004                                                                                                                                                  | Female registered nurse, male registered professionals 71,887 (13,943/57,944)            | 68.6/30.2%                                                 | PPIs                                                              | Median 13.8 years    | PPI 2033/13,943 (14.6%), nonuser 20,092/57,944 (34.7%) | Race, smoking, BMI, physical activity, Alternate Healthy Eating Index-2010, alcohol, NSAID, H2RA, cancer, MI, stroke, hypertension, diabetes mellitus, hypercholesterolemia, gastroesophageal reflux disease, Barrett's esophagus, PUD, gastrointestinal bleeding, COPD                                                                                                                        | aHR 1.19 (1.13–1.24)       |
| Maggio et al., 2013 [36]       | Italy             | Cohort       | Acute care medical awards, long-term care, rehabilitation units/2007                                                                                                                              | Older patients discharged from acute care hospitals 491 (174/317)                        | 80.0 (5.9)/46.0%                                           | Omeprazole, pantoprazole, lansoprazole, rabeprazole, esomeprazole | 1 year               | PPI 32/174 (18.4%)                                     | Age, sex, BMI, hypoalbuminemia, cognitive impairment, depression, dependency in activities of daily living, cumulative illness rating scale, peptic ulcer, diarrhea, infectious disease, fracture, the number of drugs at discharge, antithrombotic use, NSAIDs                                                                                                                                | aHR 1.51 (1.03–2.27)       |

Table 1 (continued)

| Author, year                       | Country   | Study design | Data source/study period                                                         | Definition and number of participants (PPIs/nonusers)                                                            | Mean age (SD)/male                                               | Exposure/co-medication | Follow-up        | Numbers of death                                    | Adjustments (matching variables in case-control studies)                                                                                                                                                                                                                                                                                                                                               | Mortality (95% CI)     |
|------------------------------------|-----------|--------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------|------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Mahabaleshwarkar et al., 2013 [37] | USA       | Case-control | Medicare beneficiaries /2006–2008                                                | Patients aged $\geq 65$ years receiving clopidogrel 43,159 (15,415/28,104)                                       | 76.8/38.5%                                                       | PPIs/clopidogrel       | Mean 443.1 days  | NR                                                  | Sex, race, ESRD, dual eligibility, angina, COPD, CHF, diabetes, dialysis, erosive esophageal reflux disease, GI bleeding, history of AML, CABG, PCR, stroke, hyperlipidemia, hypertension, PVD, CYP2C19 inhibitors, CYP2A4 inhibitors, ACEI, beta-adrenergic antagonists, calcium channel antagonists, diuretics, statins, sulfonylureas, aspirin, digoxin, spironolactone (age, time to cohort entry) | aOR 1.40 (1.29–1.53)   |
| Maret-Ouda et al., 2022 [38]       | Sweden    | Cohort       | Swedish Prescribed Drugs and Health Cohort/2005–2019                             | Patients who received clopidogrel after primary percutaneous coronary intervention 99,836 (35,772/64,064)        | PPI median 69 (IQR 61–77)<br>Nonuser median 67 (IQR 59–75)/72.2% | PPIs/clopidogrel       | Up to 12 months  | PPI 1434/36,110 (4.0%), nonusers 1074/64,064 (1.7%) | Age, sex, calendar year, obesity-related diseases, tobacco-related diseases, hypertension, CHF, CCI                                                                                                                                                                                                                                                                                                    | aHR 1.71 (1.58–1.86)   |
| Muñoz-Torero et al., 2020 [41]     | Spain     | Cohort       | The Factores de Riesgo y Enfermedad Arterial (FRENA) Registry/2003–2016          | Patients with coronary, cerebrovascular, or peripheral artery disease 5170 (2289/2881)                           | PPI 67.30 (11.69)<br>Nonusers 65.96 (12.04)/73.9%                | PPIs                   | Median 36 months | PPI 135/2289 (5.9%), nonuser 132/2881 (4.6%)        | Age, sex, BMI, cancer, chronic lung disease, heart failure, atrial fibrillation, hypertension, diabetes, smoking, creatinine clearance, cholesterol, CCB, diuretics, antiplatelets, insulin, statins, antidiabetics, beta-blockers                                                                                                                                                                     | aHR 1.37 (1.04–1.79)   |
| Nayan et al., 2018 [42]            | Canada    | Cohort       | Canadian Institute for Health Information Discharge Abstract Database/1997–2013  | Patients aged $\geq 65$ with newly diagnosed kidney cancer 9124 (4651/4473)                                      | 73.8 (6.0)/58.7%                                                 | PPIs                   | 60 months        | NR                                                  | Age, sex, comorbidity score, disease stage, socioeconomic status, rurality, year of diagnosis                                                                                                                                                                                                                                                                                                          | aHR 0.95 (0.92–0.98)   |
| Oudit et al., 2011 [43]            | Canada    | Cohort       | linked administrative databases/1999–2005                                        | Patients aged $\geq 65$ with heart failure 22,107 (6431/15,676)                                                  | Median 80 years (IQR 74–86)/48%                                  | PPIs                   | 1 year           | PPI 1153/6431 (26%), nonuser 5659/15,676 (32%)      | Sex, age, hypertension, diabetes, IHD, CVD, PVD, COPD, renal disease, cancer, anemia, GI bleeding, cardiac medications, ACEI, ARB, beta-blockers, spironolactone, digoxin, statin, CCB, non-cardiac medications, sulfonyleurea                                                                                                                                                                         | aOR 0.87 (0.81–0.93)   |
| Pani et al., 2020 [44]             | Italy     | Cohort       | Internal Medicine Units of the Niguarda Hospital/2015–2018                       | Patients admitted to internal medicine ward 531 (231/300)                                                        | 77.3 (13.1)/60.3%                                                | PPIs                   | NA               | PPI 22/231 (9.5%), nonuser 31/300 (10.3%)           | -                                                                                                                                                                                                                                                                                                                                                                                                      | OR 0.940 (0.527–1.678) |
| Pegoli et al., 2017 [45]           | Australia | Cohort       | Older hospital patients discharged to residential aged care facilities/2014–2015 | Patients $\geq 75$ years and discharged from a tertiary hospital to residential aged care facilities 102 (51/51) | 87 (6)/59.2%                                                     | PPIs                   | 6 months         | PPI 19/51 (37.3%), nonuser 14/51 (27.5%)            | Age, gender, initial place of residence, CCI, total number of days spent in hospital within 6 months, number of regular medications taken on discharge                                                                                                                                                                                                                                                 | aHR 2.24 (1.01–4.96)   |

Table 1 (continued)

| Author, year                         | Country     | Study design | Data source/study period                                                                                            | Definition and number of participants (PPIs/nonusers)                                                                                                      | Mean age (SD)/male                                                                                                              | Exposure/co-medication                                                             | Follow-up                                 | Numbers of death                                                                                                                                        | Adjustments (matching variables in case-control studies)                                                                                                                                                                                                                                                                                                                   | Mortality (95% CI)                                                            |
|--------------------------------------|-------------|--------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Rassen et al., 2009 [46]             | Canada, USA | Cohort       | Health insurance program (British Columbia, Pennsylvania, New Jersey)/2001–2005                                     | Patients aged $\geq 65$ years and who underwent percutaneous coronary intervention or were hospitalized for acute coronary 18,565 (3996/14,569)            | British Columbia 74 (range 65–100), Pennsylvania 78 (range 65–101), New Jersey 78 (range 65–104)/76.1%                          | PPIs                                                                               | Median 29–30 days                         | PPI 61/3996 (1.5%), nonuser 124/14,569 (0.9%)                                                                                                           | Age, sex, race, calendar year, outpatient visit, PCI performed, inpatient, MI, gastrointestinal bleed, angina, hemorrhagic stroke, nonselective NSAID, COX-2 inhibitor, diabetes drug, statin, beta-blocker, ACEI/ARB, warfarin, history of diabetes, hypertension, PVD, CHF, nursing home residence, number of medications, time in hospital, office visits, CCI          | aHR 1.20 (CI 0.84–1.70)                                                       |
| Roberts et al., 2021 [47]            | Australia   | Cohort       | Five major public hospitals/2017                                                                                    | Patients aged $\geq 75$ years, admitted to general medical wards, discharged to a RACF and receiving $\geq 5$ regular medications at admission 170 (95/75) | 86.7 (6.9)/44.7%                                                                                                                | PPIs/ $\geq 5$ regular medications                                                 | NR                                        | NR                                                                                                                                                      | Residence, CCI, number of medications                                                                                                                                                                                                                                                                                                                                      | aHR 1.09 (0.63–1.89)                                                          |
| Rooney et al., 2021 [48]             | USA         | Cohort       | Four US communities (For-syth County, NC; Jackson, MS; suburbs of Minneapolis, MN; Washington County, MD)/2011–2017 | Participants who have atherosclerosis Risk without prevalent cardiovascular disease 4436 (1067/3369)                                                       | PPI 75.3 (5.1), nonuser 75.2 (5.1)/56.8%                                                                                        | PPIs                                                                               | Median 5.6 years                          | PPI 39/1067 (3.7%), nonuser 82/3369 (2.4%)                                                                                                              | Age, race-center, sex, education, smoking status, drinking status, physical activity, obesity, diabetes, systolic blood pressure, antihypertension medication use, lipid-lowering medication use, HDL, total cholesterol, eGFR, hypomagnesemia                                                                                                                             | aHR 1.34 (0.86–2.09)                                                          |
| Sharma et al., 2019 [11]             | USA         | Cohort       | National linked SEER-Medicare databases/2007–2012                                                                   | Patients aged $\geq 66$ years and received tyrosine kinase inhibitors 12,538 (9695/2842)                                                                   | Median 76 years/48.7%                                                                                                           | PPIs/TKIs                                                                          | 1 year                                    | PPI 5745/9695 (59.3%), nonusers 1600/2842 (56.3%)                                                                                                       | Age, sex, race, marital status, region, primary cancer type, stage at diagnosis, receipt of radiation, score on Deyo adaptation of the CCI, year of TKI start, polypharmacy, prior PPI use within 90 days before the TKI start date, receipt of prior chemotherapy, NSAID, history of gastric ulcer, duodenal ulcers, time between cancer diagnosis and the TKI start date | aHR 1.10 (1.04–1.17)                                                          |
| Sturm et al., 2021 [49]              | Germany     | Cohort       | University Medical Center Freiburg, Germany/2009–2019                                                               | Patients with hepatocellular carcinoma treated by transarterial chemoembolization 358 (167/191)                                                            | Median 69 (IQR 60–75)/83.0%                                                                                                     | PPIs                                                                               | Median transplant-free survival 22 months | PPI 15/167 (9.0%), nonusers 15/191 (7.9%)                                                                                                               | Age, sex, Child Pugh stadium, Solitary vs multifocal tumor, tumor diameter, portal vein thrombosis, Serum albumin, ferritin, PPI dose                                                                                                                                                                                                                                      | aHR 1.53 (1.13–2.06)                                                          |
| Tearamura-Grönblad et al., 2012 [50] | Finland     | Cohort       | 1. Assisted living facilities/2007<br>2. Long-term care hospitals/2003<br>3. Nursing homes, acute geriatric wards   | Institutionalized older people 1. 1389 (367/1022)<br>2. 1004 (231/773)<br>3. 425 (91/334)                                                                  | 1. PPIs 83.6 (7.7), nonusers 82.4 (7.8)<br>2. PPIs 80.6 (11.4), nonusers 81.5 (10.8)<br>3. PPIs 85.5 (7.4), nonusers 86.3 (6.8) | Omeprazole, pantoprazole, lansoprazole, rabeprazole, esomeprazole (ATC code A02BC) | 1 year                                    | 1. PPIs 74/367 (20.2%), nonusers 208/1022 (20.4%)<br>2. PPIs 77/231 (33.3%), nonusers 206/773 (26.6%)<br>3. PPIs 33/91 (36.3%), nonusers 73/334 (21.9%) | 1. Age, sex, CCI, immobility, SSRIs<br>2. Age, sex, CCI, SSRIs, malnutrition<br>3. Age, sex, CCI, delirium, aspirin, SSRIs                                                                                                                                                                                                                                                 | 1. aHR 1.06 (0.77–1.46)<br>2. aHR 1.36 (1.04–1.77)<br>3. aHR 1.90 (1.23–2.94) |



Table 1 (continued)

| Author, year               | Country   | Study design | Data source/study period                                                                                                  | Definition and number of participants (PPIs/nonusers)                        | Mean age (SD)/male                              | Exposure/co-medication | Follow-up                 | Numbers of death                                      | Adjustments (matching variables in case-control studies)                                                                                                                                                                                                                                   | Mortality (95% CI)    |
|----------------------------|-----------|--------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------|------------------------|---------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Tomisaki et al., 2022 [51] | Japan     | Cohort       | Hospital of the University of Occupational and Environmental Health/2018–2021                                             | Patients with advanced urothelial carcinoma 40 (15/25)                       | 72 (IQR 67–79)/75%                              | PPIs/pembrolizumab     | Median 11.2 months        | NR                                                    | Age, gender, ECOG-PS, hemoglobin, liver metastasis, time from prior chemotherapy, antibiotics                                                                                                                                                                                              | aHR 4.00 (1.22–13.15) |
| Tran et al., 2021 [52]     | UK        | Cohort       | United Kingdom Clinical Practice Research Datalink, Hospital Episode Statistics, Office for National Statistics/2003–2017 | Patients with idiopathic pulmonary fibrosis 3036 (1852/1852)                 | PPI 75.4 (9.4), nonusers 75.6 (9.5)/77.7%       | PPIs                   | Median survival 2.8 years | PPI 1221/1852 (66.0%), nonusers 482/1852 (26.0%)      | Age, sex, BMI, smoking, excessive alcohol use, ethnicity, prior hospitalizations, comorbidities, asthma, COPD, GERD, Nissen fundoplication, CVD, diabetes, cancer, renal disease, depression, corticosteroids, ACEI, beta-blockers, anticoagulants, diuretics, statins, NSAID, H2-blockers | aHR 1.07 (0.94–1.22)  |
| Tsai et al., 2021 [53]     | Taiwan    | Cohort       | Taiwan's National Health Insurance Research Database/1999–2011                                                            | Patients with dialysis 18,230 (9115/9115)                                    | PPI 66.59 (14.82), nonusers 66.58 (14.82)/57.8% | PPIs                   | 3.52 years                | PPI 4660/9115 (51.1%), nonusers 4449/9115 (48.8%)     | Age, sex, peptic ulcer, duodenal ulcer, gastric ulcer, liver cirrhosis, CKD, acute on chronic kidney disease after index discharge, propensity covariates                                                                                                                                  | aHR 1.03 (0.98–1.08)  |
| Wang et al., 2021 [54]     | Sweden    | Cohort       | Swedish Prescribed Drug Registry, Swedish Causes of Death Registry/2006–2013                                              | Patients with colorectal cancer 32,411 (17,801/14,610)                       | > 65 years 71%/52%                              | PPIs                   | Median 2.74 years         | PPI 4746/17,801 (26.7%), nonusers 8439/14,610 (57.8%) | Age, sex, pre-diagnosis PPI use, tumor site, tumor stage, surgery, low-dose aspirin, non-aspirin NSAIDs, CCI                                                                                                                                                                               | aHR 1.38 (1.32–1.44)  |
| Wilson et al., 2011 [55]   | Australia | Cohort       | Intermediate-level residential aged care facilities                                                                       | Institutionalized older people 602 (246/356)                                 | 85.7 (6.4)/29.1%                                | PPIs                   | 1 year                    | PPIs 27/246 (11.0%), nonusers 35/356 (9.8%)           | Age, sex, MCCI                                                                                                                                                                                                                                                                             | aHR 1.08 (0.63–1.86)  |
| Xie et al., 2022 [57]      | China     | Cohort       | Medical Information Mart for Intensive Care/2011–2019                                                                     | Patients with acute myocardial infarction and admitted to ED 2001 (426/1575) | 69.6 (14.2)/59.2%                               | PPIs                   | NR                        | PPI 18/426 (4.2%), nonusers 86/1575 (5.5%)            | Age, sex, ethnicity, troponin T, creatinine kinase, PTCA, dilation of coronary artery, CCI, CHF, cerebrovascular disease, PVD, dementia, chronic pulmonary disease, mild liver disease, renal disease                                                                                      | aHR 1.08 (0.58–1.99)  |

ACEI angiotensin-converting enzyme inhibitor, aHR adjusted hazard ratio, aOR adjusted odds ratio, APRI Platelet Ratio Index, ARB angiotensin receptor blocker, aRR adjusted risk ratio, ATC anatomical therapeutic chemical classification, BMI body mass index, CAD coronary artery disease, CCB calcium channel blockers, CCI Charlson comorbidity index, CHF congestive heart failure, CI confidence interval, COPD chronic obstructive pulmonary disease, CVD cardiovascular disease, EAC esophageal adenocarcinoma, ECOG Eastern Cooperative Oncology Group, eGFR estimated glomerular filtration rate, ESCC esophageal squamous cell carcinoma, GCC cardia gastric cancer, GERD gastroesophageal reflux disease, GI gastrointestinal, GNCC non-cardia gastric cancer, HBV hepatitis B virus, HCC hepatocellular carcinoma, HCV hepatitis C virus, HR hazard ratio, ICI immune checkpoint inhibitor, IHD ischemic heart disease, MCCI medical-based Charlson comorbidity index, MELD Model for End-Stage Liver Disease, MI myocardial infarction, NA not applicable, NAFLD nonalcoholic fatty liver disease, NOAC non-vitamin K antagonist oral anticoagulants, NR not reported, NSCLC nonsmall cell lung cancer, NSAID nonsteroidal anti-inflammatory drug, OAC oral anticoagulants, PPI proton pump inhibitor, PTCA percutaneous transluminal coronary angioplasty, PUD peptic ulcer disease, PVD peripheral vascular disease, RCT randomized controlled trial, SEER Surveillance, Epidemiology, and End Results, TACE transarterial chemoembolization, TKI tyrosine kinase inhibitors, VTE venous thromboembolism



were included in the meta-analysis when we could calculate RRs from the data on OR and the incidence of mortality in the control group. In addition, we performed subgroup and sensitivity analyses according to follow-up period, population, country, study design, the median age of included studies, and quality of the studies. We conducted a duration analysis of the risk of death among PPI users by follow-up period:  $\leq 6$  months,  $> 6$  months– $\leq 1$  year,  $> 1$ – $\leq 2$  years,  $> 2$ – $\leq 3$  years,  $> 3$ – $\leq 4$  years, and  $> 4$  years. To further evaluate disease-specific mortality, we evaluated the association between PPIs and mortality among patients with cancers, cardiovascular disease, and kidney diseases as well as people who were institutionalized and hospitalized. To assess the impact of different health care systems, the subgroup analysis by countries was conducted: USA/Canada, Europe, Australia, and Asia. The association of subgroups was also examined by patient's age to take into account differences in mortality by age groups ( $\leq 75$ , 76–85, and  $> 85$  years old) as well as by quality assessment of included studies (low, moderate, and serious/critical risk of bias) to incorporate assessment into the analyses. We also analyzed the association in studies that included and did not include Charlson Comorbidity Index (CCI) as a covariate.

Heterogeneity was assessed using the  $I^2$  test and  $Q$  statistic, with the significance of the  $Q$ -statistic test being considered at  $p < 0.05$ . Heterogeneity was considered for  $I^2$  values of more than 50% [13]. The funnel plot was used to estimate possible publication bias owing to the tendency to publish studies with positive results. We used Review Manager 5.3 software (The Nordic Cochrane Centre, The Cochrane Collaboration, 2014).

## Results

### Literature search

An initial search was performed on 15,202 articles; we identified 13,544 relevant ones after excluding 1658 duplicate articles (Fig. 1). In the screening process, 12,992 articles were removed during the title/abstract review, and 511 articles were excluded during the full-text review due to one of the following reasons: no elderly patients, no PPI therapy, ineligible study design, no comparator group available, no outcomes of interest reported, and non-original studies. Finally, 49 cohorts among 41 studies (four RCTs and 37 observational studies) were included in our systematic review [11, 18–57] since five studies included more than one cohort in their studies [19, 20, 23, 26, 50]. In the meta-analysis, 4 cohorts among 4 RCTs and 44 cohorts among 36 observational studies were included. One

study/cohort was not included in the meta-analysis as we could not calculate RR from OR due to a lack of data on the number of deaths [29].

### General characteristics of the included studies

We included four RCTs, 34 cohort studies, and three case–control studies (Table 1). In RCTs, PPIs including omeprazole and pantoprazole were included as interventions with comparators of placebo. In observational studies, most studies included all kinds of PPIs as interventions and non-PPI users as controls including H2RA. The studies were conducted in the USA, Canada, Europe, Australia, and Asia. The study participants included the following: patients with cancer ( $n = 9$ ), cardiovascular disease ( $n = 11$ ), kidney disease ( $n = 4$ ), institutionalization ( $n = 6$ ), and hospitalization ( $n = 5$ ). All studies included 2,515,079 participants with a median age ranging from 67 to 96 years and the percentage of males ranging from 18.4 to 99.5%. Study follow-up periods were from 21 days to 13.8 years. When estimating adjusted HRs or ORs, baseline demographics, disease-related clinical factors, comorbidities, and medications were included as confounding variables (Table 1).

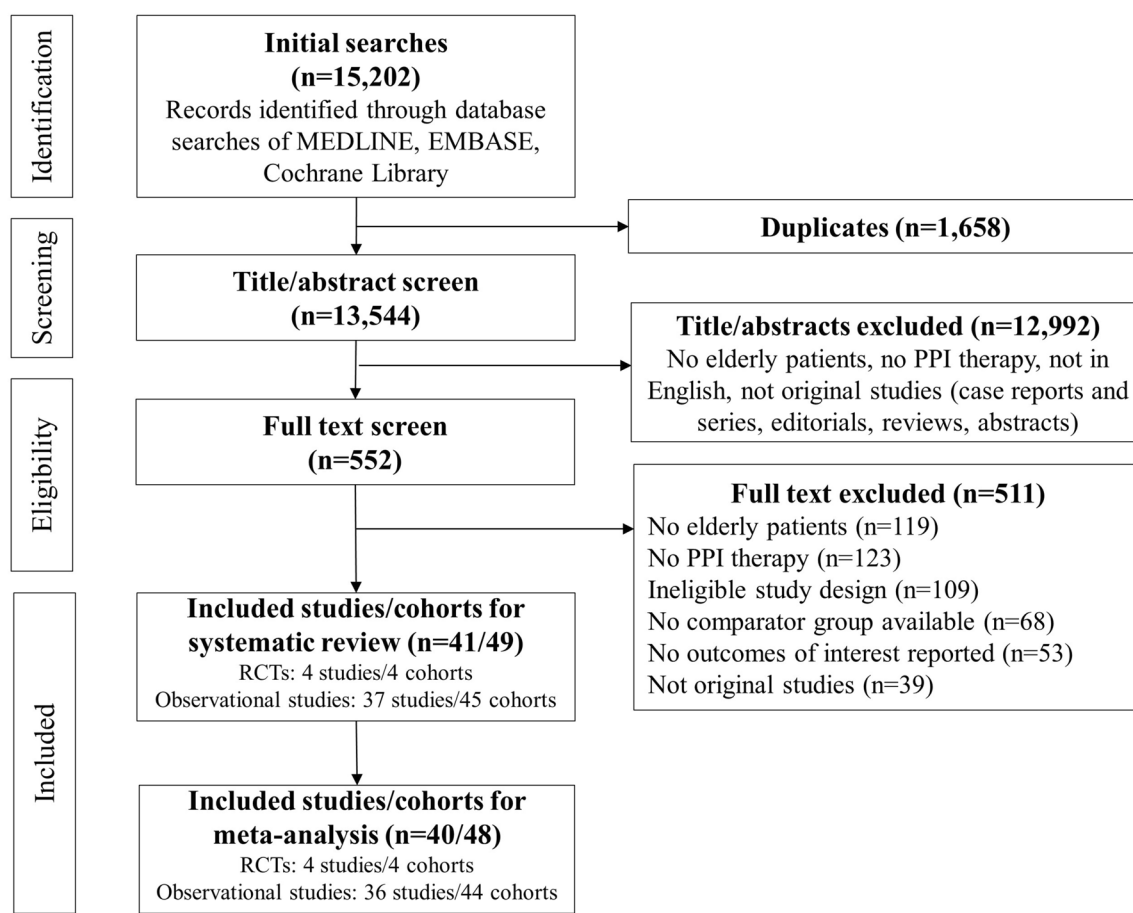
### Quality assessment

The included RCTs had a low risk of bias from missing outcome data, measurement of the outcome, and selection of the reported result (Supplement Fig. 1). For measurement of the outcome, we determined a low risk because mortality was not affected by any measurement methods. We evaluated some concerns for deviations from intended interventions for half of the included RCTs because PPIs can also be used as over-the-counter (OTC) drugs.

All cohort studies had a low risk of bias from deviations from intended interventions, missing data, measurement of the outcome, and selection of the reported result (Supplement Table 2). Among 37 cohort studies, 10 studies had moderate, serious, or critical risk of bias because appropriate confounding covariates for the adjusted estimates were not included, and 19 had a moderate risk for classification of interventions because OTC use of PPIs could not be captured. The overall risk of bias was assessed as low at 27%, moderate at 57%, and serious or critical risk at 16%.

### Proton pump inhibitors and mortality in RCTs

Among 4 RCTs, the death rates were 11.8% in PPI users (1286/10,918) and 11.4% in nonusers (1250/10,928). The PPI use and risk of mortality in unadjusted RR among



**Fig. 1** PRISMA flow diagram of study selections

individuals aged 50 years or older were not significantly associated (RR 1.03; 95% CI, 0.91–1.16), with moderate heterogeneity was moderate ( $I^2 = 45\%$ ,  $p = 0.14$  (Fig. 2A). When estimating adjusted risk ratio among 20,859 elderly people from 2 RCTs, the association between PPI users and death was not significantly associated compared to nonusers (aRR, 1.01; 95% CI, 0.94–1.08), with 0% of heterogeneity ( $I^2 = 0\%$ ,  $p = 0.60$ ) (Fig. 2C).

### Proton pump inhibitors and mortality in observational studies

In 26 observational studies (29 cohorts), there were 75,675 deaths in 512,263 PPI users (14.8%) and 94,428 deaths in 1,731,521 nonusers (4.9%). We found no significant association between the use of PPI and mortality from unadjusted RR in the elderly (RR 1.14; 95% CI, 0.90–1.45) (Fig. 2B). There was significant heterogeneity ( $I^2 = 100\%$ ,  $p < 0.001$ ), and funnel plots showed no evidence of publication bias (Fig. 3A). For adjusted

estimates, 41 cohorts from 33 studies of the association between PPIs and death included 2,429,961 individuals aged 50 years or older. The use of PPI was significantly associated with a 15% increased risk of mortality compared to non-use (aRR, 1.15; 95% CI, 1.10–1.20) (Fig. 2D). Significant heterogeneity was detected ( $I^2 = 93\%$ ,  $p < 0.001$ ), and there was no evidence of publication bias based on the funnel plot (Fig. 3B).

Additionally, we conducted subgroup and sensitivity analyses of adjusted risk ratios among observational studies. PPI use among the elderly was significantly associated with an increased risk of mortality (by 17–32%) compared to non-use in the follow-up period between more than 6 months and less than 4 years, while the association was not significant in a shorter follow-up of a less than 6 months (aRR, 1.02; 95% CI, 0.45–2.33) and  $\geq 4$ -year follow-up (aRR, 1.06; 95% CI, 0.99–1.14) (Table 2). The mortality risk of PPI users among patients with cancer, cardiovascular diseases, kidney diseases, and institutionalization showed a similar trend. When investigating the risk of death by country, the association between PPIs

**Table 2** Subgroup analysis of the association between proton pump inhibitors and the adjusted risk of mortality of observational studies

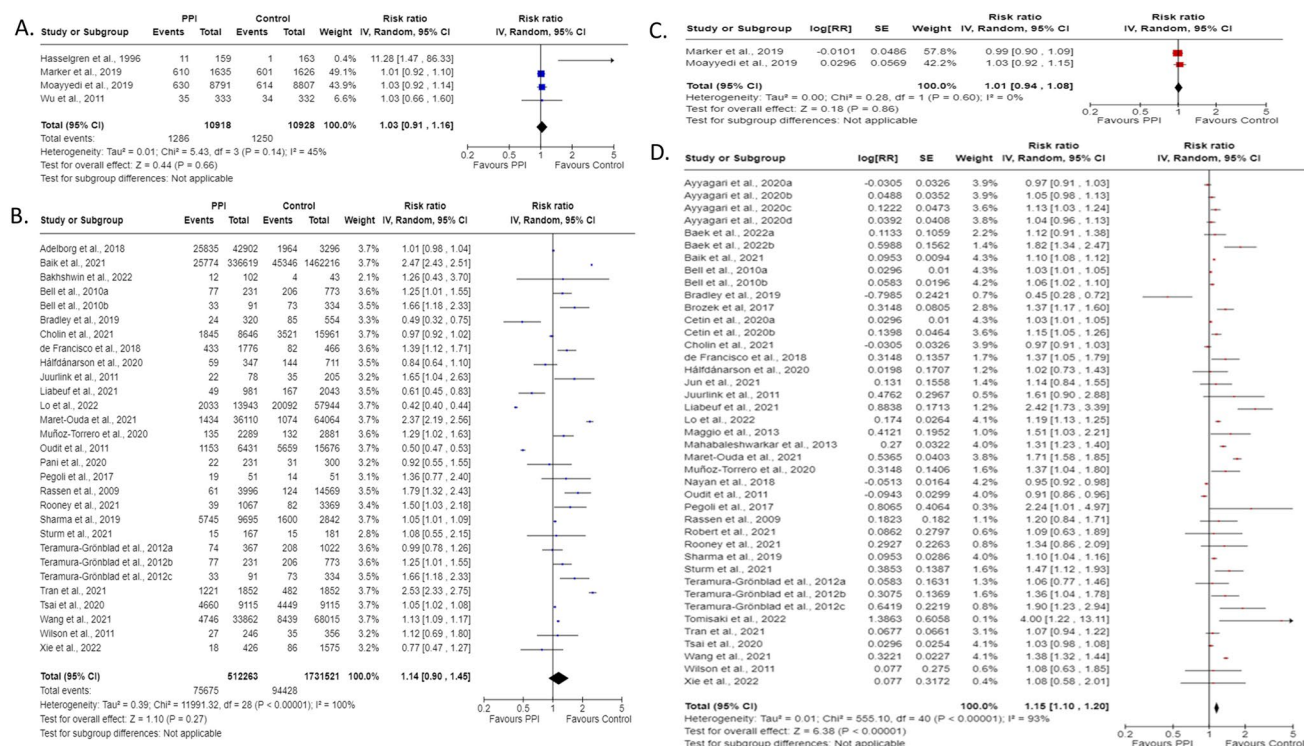
| Observational studies                           | Cohort, <i>n</i> | PPI, <i>n</i> | Control, <i>n</i> | Random effects, risk ratio (95% CI) | Effect, <i>P</i> -value | <i>I</i> <sup>2</sup> | Heterogeneity, <i>P</i> -value |
|-------------------------------------------------|------------------|---------------|-------------------|-------------------------------------|-------------------------|-----------------------|--------------------------------|
| <b>Follow-up</b>                                |                  |               |                   |                                     |                         |                       |                                |
| ≤ 6 months                                      | 3                | 4367          | 15,174            | 1.02 (0.45–2.33)                    | 0.960                   | 87%                   | < 0.001                        |
| > 6 months– ≤ 1 year                            | 12               | 53,429        | 86,745            | 1.19 (1.07–1.33)                    | 0.001                   | 94%                   | < 0.001                        |
| > 1 year– ≤ 2 years                             | 3                | 17,358        | 28,761            | 1.32 (1.24–1.40)                    | < 0.001                 | 0%                    | 0.690                          |
| > 2 years– ≤ 3 years                            | 4                | 33,332        | 23,911            | 1.28 (1.12–1.49)                    | < 0.001                 | 77%                   | 0.004                          |
| > 3 years– ≤ 4 years                            | 3                | 346,715       | 1,533,230         | 1.17 (1.03–1.32)                    | 0.020                   | 93%                   | < 0.001                        |
| > 4 years                                       | 8                | 63,677        | 133,020           | 1.06 (0.99–1.14)                    | 0.100                   | 89%                   | < 0.001                        |
| <b>Patients</b>                                 |                  |               |                   |                                     |                         |                       |                                |
| Cancers <sup>a</sup>                            | 13               | 41,271        | 29,835            | 1.15 (1.03–1.27)                    | 0.009                   | 94%                   | < 0.001                        |
| Cardiovascular diseases                         | 8                | 65,492        | 130,443           | 1.29 (1.01–1.64)                    | 0.040                   | 97%                   | < 0.001                        |
| Kidney diseases                                 | 4                | 20,518        | 27,584            | 1.20 (1.02–1.42)                    | 0.030                   | 88%                   | < 0.001                        |
| Institutionalization                            | 7                | 1308          | 3643              | 1.08 (1.01–1.16)                    | 0.020                   | 64%                   | 0.010                          |
| Hospitalization                                 | 2                | 269           | 392               | 1.36 (0.99–1.86)                    | 0.060                   | 0%                    | 0.340                          |
| <b>Countries</b>                                |                  |               |                   |                                     |                         |                       |                                |
| USA/Canada                                      | 15               | 409,389       | 1,672,520         | 1.06 (0.99–1.12)                    | 0.070                   | 93%                   | < 0.001                        |
| Europe                                          | 17               | 109,963       | 146,758           | 1.27 (1.18–1.38)                    | < 0.001                 | 95%                   | < 0.001                        |
| Australia                                       | 3                | 392           | 482               | 1.27 (0.85–1.89)                    | 0.250                   | 23%                   | 0.280                          |
| Asia <sup>a</sup>                               | 5                | 9556          | 10,715            | 1.28 (0.99–1.66)                    | 0.060                   | 78%                   | 0.001                          |
| <b>Study design</b>                             |                  |               |                   |                                     |                         |                       |                                |
| Cohort <sup>a</sup>                             | 39               | 512,398       | 1,800,932         | 1.14 (1.09–1.20)                    | < 0.001                 | 93%                   | < 0.001                        |
| Case-control                                    | 2                | 15,510        | 28,309            | 1.31 (1.23–1.40)                    | < 0.001                 | 0%                    | 0.490                          |
| <b>Median age</b>                               |                  |               |                   |                                     |                         |                       |                                |
| ≤ 75 years old <sup>a</sup>                     | 20               | 89,642        | 171,648           | 1.17 (1.08–1.27)                    | < 0.001                 | 93%                   | < 0.001                        |
| 76–85 years old                                 | 13               | 72,129        | 115,548           | 1.09 (1.02–1.16)                    | 0.007                   | 88%                   | < 0.001                        |
| > 85 years old                                  | 4                | 328           | 794               | 1.38 (0.95–2.00)                    | 0.090                   | 71%                   | 0.020                          |
| <b>ROBINS-I</b>                                 |                  |               |                   |                                     |                         |                       |                                |
| Low <sup>a</sup>                                | 8                | 15,845        | 36,801            | 1.13 (1.01–1.28)                    | 0.040                   | 82%                   | < 0.001                        |
| Moderate                                        | 22               | 455,643       | 1,715,161         | 1.17 (1.10–1.24)                    | < 0.001                 | 92%                   | < 0.001                        |
| Serious/critical                                | 5                | 12,043        | 6260              | 1.08 (1.02–1.15)                    | 0.010                   | 71%                   | 0.007                          |
| <b>CCI</b>                                      |                  |               |                   |                                     |                         |                       |                                |
| Include CCI as a covariate                      | 16               | 71,285        | 102,594           | 1.22 (1.09–1.36)                    | < 0.001                 | 95%                   | < 0.001                        |
| Did not include CCI as a covariate <sup>a</sup> | 25               | 469,836       | 1,748,500         | 1.11 (1.06–1.16)                    | < 0.001                 | 90%                   | < 0.001                        |

CCI Charlson comorbidity index, PPI proton pump inhibitor, RCT randomized controlled trial, ROBINS-I Risk of Bias Assessment tool for Non-randomized Studies of Interventions, USA United States

<sup>a</sup>Baek et al. did not report the number of PPIs and nonusers of patients aged 65 years or older

and increased mortality was significant among studies conducted in Europe (aRR, 1.27; 95% CI, 1.18–1.38), while the association was towards significant among studies conducted in USA/Canada, Asia, and Australia. The significant association between PPI use and mortality appeared consistent among cohort studies (aRR, 1.14; 95% CI, 1.09–1.20) and case–control studies (aRR, 1.31; 95% CI, 1.23–1.40). According to the median age of participants in the included studies, the results of the association

between PPI use and mortality remain similar to the base-case analysis results. The lower the risk of bias of the included cohort studies, the larger the RR (aRR, 1.13; 95% CI, 1.01–1.28 in low risk of bias vs aRR, 1.08; 95% CI, 1.02–1.15 in serious/critical risk of bias). The association of risk of death and PPI users among both studies included CCI as a covariate and studies that did not include CCI as a covariate remained significant.

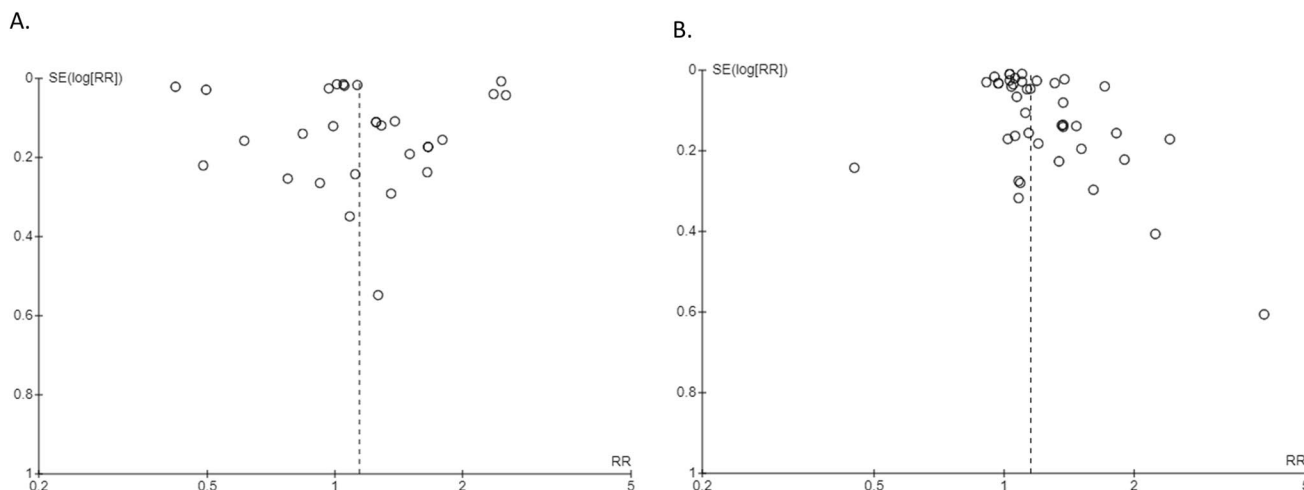


**Fig. 2** The association between proton pump inhibitors and mortality. **A** Unadjusted risk ratio in randomized controlled trials. **B** Unadjusted risk ratio in observational studies. **C** Adjusted risk ratio in randomized controlled trials. **D** Adjusted risk ratio in observational studies

## Discussion

This systematic review found a 15% increase in the risk of death among elderly people taking PPI compared to those not taking PPI. In particular, the association between PPI use and mortality was significant in studies among senior adults with cancer, cardiovascular diseases, and kidney diseases as well as studies with longer follow-ups and cohort studies.

Our results are aligned with those of previous studies that included not only adults but also elderly [9, 58]. A previous study using administrative data from Veterans Affairs demonstrated an excess risk of death among adults taking PPIs with a mean age of 61 years [9]. A systematic review also showed that all-cause mortality increased in adults treated with PPI, especially older adults treated with long-term PPI, compared to patients not treated with PPI [58]. Although a



**Fig. 3** Funnel plot of observational studies. **A** Unadjusted risk ratio. **B** Adjusted risk ratio

clear mechanism is not known for the association between PPIs and the risk of death, a potential biological mechanism was suggested. Prolonged exposure to PPI impairs endothelial function, increases oxidative stress, slows lysosomal acidification and protein accumulation in endothelial cells, and accelerates human endothelial aging by shortening telomere length [59]. PPIs also upregulate the expression of protein levels and mRNA resulting in increased activity of the heme oxygenase-1 enzyme in gastric and endothelial cells, which reduces beneficial effects, including antiapoptotic, anti-inflammatory, antioxidant, antiproliferative, and immunomodulatory effects in vascular cells [9, 60, 61]. Elderly patients may be more vulnerable to side effects during long-term PPI use on account of their comorbidities, use of multiple drugs, and poor nutrition [36]. PPIs have also been significantly overused in the elderly, contrary to the recommendation to restrict the PPI treatment period to fewer than 12 weeks [58].

Our subgroup and sensitivity analyses demonstrated that PPI use significantly affected the increased death among older patients with cancer, cardiovascular disease, and kidney disease. These results are similar to those of previous studies that were not limited to the elderly, which correlated PPI usage with excess mortality from cardiovascular disease, chronic kidney disease, and upper gastrointestinal and lung cancers [10, 62, 63] and also suggested an association between PPI usage and increased risk of the aforementioned diseases [62, 64]. The authors suggested a mechanism leading to the excess cause-specific mortality related to PPI use is linked to the exacerbation of underlying disease or the development of a new disease that increases the risk of cause-specific death. This mechanism was also suggested as a unifying mechanism [59]. In addition, inappropriate prescription of high-dose PPI during or after hospitalization is frequent as antithrombotic agents are widely used in elderly patients [36].

Notably, in cohort studies and studies with long-term follow-up (more than 1 year to 3 years), 28–32% of increased risk of death was clearly observed. However, this was not observed when the follow-up period was shortened (less than 6 months). A previous review also showed that there was no immediate apparent increase in all-cause mortality in adults taking PPIs in one RCT, while increased mortality with PPI use was observed in the observational studies followed up to 1 year [58]. Although RCTs have a higher hierarchy of an evidence-based approach than observational studies because of randomization, the findings from observational studies can show the results with all confounders, including unidentified confounders in the real-world population [58]. Our systematic review showed that significant results were obtained in terms of real-world evidence rather than in RCT settings, suggesting that real-world evidence should be

considered when applied to clinical settings for patient care, such as pharmacovigilance criteria.

Our systematic review has strengths that, to our best knowledge, this is the first meta-analysis study to explore the association between PPI use and mortality in the older adult population with a large sample size. The significance of the pooled estimates was higher in high-quality studies with a low overall risk of bias. Our diverse subgroup results categorized by the follow-up period, disease, study design, country, and risk of bias provide comprehensive and detailed information. Notably, our results clearly showed that PPI use was significantly associated with increased death in older patients with cancer, cardiovascular disease, and kidney disease. In addition, we provide real-world evidence based on results from cohort studies and long-term follow-up studies.

Our study has several limitations. Primarily, we were not able to investigate the impact of mortality by the duration or dosage of PPI use due to a lack of data. However, our best available evidence supports the association the PPIs and mortality in elderly people. Secondly, the level of heterogeneity was low in RCTs, and it may be caused by study design or the small number of included studies. However, the pooled estimates in observational studies have significant heterogeneity although we conducted several subgroup analyses to find the cause. It may be caused by various populations, different levels of controlling covariates, and unknown factors. Despite the diversity of included studies, results were consistent across subgroups. Thirdly, half of the cohort studies in the meta-analysis showed a moderate risk of bias in the classification of intervention due to self-administrated OTC without prescriptions in most countries. However, the significant association between PPI use and mortality remained consistent in studies with low, moderate, and serious/critical risk of bias. Furthermore, the relative estimates can be considered conservative as individuals who may take OTCs may be included as both PPI users and nonusers. Finally, we cannot identify the causal relationship between PPI use and the risk of mortality since most of the included studies investigated the association between PPI use and the risk of mortality. Also, there was no clear association between PPIs and death from the four RCTs in the meta-analysis.

## Conclusions

Our meta-analysis showed that the association between PPIs and the risk of mortality of RCTs was inconsistent with the association from observational studies in the elderly. The meta-analysis of RCTs showed that PPI use was not powered to detect an increased risk of death compared to nonusers. However, the increased risk of death was identified in observational studies and studies with long-term follow-ups, and



the association was consistent in the subgroup population, including among patients with cancer, cardiovascular disease, and kidney disease. Our findings highlight the need to increase awareness of increased mortality due to PPI use and to restrict PPI prescriptions to elderly people, wherein the benefits outweigh the potential risks.

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**Author contribution** HJS was involved in study concept and design, literature search, data extraction, data analysis, data interpretation, and manuscript writing. HJS was involved in data interpretation and manuscript writing. XJ was involved in literature search, data extraction, and data interpretation. NJ, YJL, and IHH were involved in literature search and data interpretation. All authors reviewed and approved the final version.

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**Data availability** The datasets generated during the current study are available from the corresponding author on reasonable request.

## Declarations

**Ethics approval** Not applicable.

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